Permutations & Combinations

ANSWER KEY

Section 1: Fundamental Counting & Factorials

1 How many 4-digit numbers can be formed using the digits $1, 2, 3, 4, 5, 6$ without repetition?

a. **360** ✓

b. 720

c. 120

d. 240

The number of permutations of 6 distinct digits taken 4 at a time is ${}^{6}P_{4} = 6 \times 5 \times 4 \times 3 = 360$.

2 In how many ways can 7 runners finish 1st, 2nd, and 3rd in a race, assuming no ties?

a. **210** ✓

b. 35

c. 5040

d. 120

${}^{7}P_{3} = 7 \times 6 \times 5 = 210$.

3 How many even 3-digit numbers can be formed using digits $1, 2, 3, 4, 5$ without repetition?

a. **24** ✓

b. 12

c. 36

d. 60

The last digit must be even: 2 choices (2 or 4). Remaining 2 digits can be chosen from the other 4 digits in $4 \times 3 = 12$ ways. Total $= 12 \times 2 = 24$.

Section 2: Permutations with Repetition & Restrictions

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4 How many distinct arrangements of all the letters in the word MATHEMATICS are possible?

a. $4,989,600$ ✓

b. $39,916,800$

c. $1,247,400$

d. $2,494,800$

11 letters total: M appears 2, A appears 2, T appears 2. Distinct permutations $= \frac{11!}{2!\,2!\,2!} = \frac{39916800}{8} = 4,989,600$.

5 In how many ways can 5 boys and 4 girls stand in a line if all 4 girls must stand together?

a. $17,280$ ✓

b. $2,880$

c. $43,200$

d. $5,760$

Treat the 4 girls as a single block: 6 items to arrange ($5! = 120$ ways). Within the block, the 4 girls can be arranged in $4! = 24$ ways. Total $= 120 \times 24 = 2880$? Wait, 5 boys + 1 block = 6 items, $6! = 720$. $720 \times 24 = 17,280$.

6 In how many ways can 6 people sit around a circular table, where rotations are considered equivalent?

a. **120** ✓

b. 720

c. 24

d. 360

Circular permutations of n items $= (n-1)!$. For 6 people: $(6-1)! = 5! = 120$.

Section 3: Combinations & Committee Selection

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7 A committee of 5 is to be chosen from a group of 8 men and 6 women. How many committees are possible if it must contain exactly 3 men and 2 women?

a. **840** ✓

b. 420

c. 1260

d. 2002

Number of ways $= \binom{8}{3} \times \binom{6}{2} = 56 \times 15 = 840$.

8 From a group of 10 students, a team of 4 is selected. If the team must include the student council president, how many distinct teams are possible?

a. **84** ✓

b. 210

c. 126

d. 72

The president is fixed, so we choose the remaining 3 team members from the remaining 9 students: $\binom{9}{3} = \frac{9 \times 8 \times 7}{6} = 84$.

9 A team of 4 is selected from 7 boys and 5 girls. How many teams have at least one girl?

a. **460** ✓

b. 495

c. 420

d. 35

Total teams without restriction $= \binom{12}{4} = 495$. Teams with no girls (all boys) $= \binom{7}{4} = 35$. Teams with at least one girl $= 495 - 35 = 460$.

Section 4: Complex Multi-Condition Problems

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10 How many 5-digit odd numbers greater than 40,000 can be formed using digits $2, 3, 4, 5, 7$ without repetition?

a. **28** ✓

b. 36

c. 48

d. 24

First digit can be 4, 5, or 7. Case 1: First digit is 4 (1 choice). Last digit must be odd from $\{3, 5, 7\}$ (3 choices). Middle 3 digits: $3! = 6$. Total $= 1 \times 3 \times 6 = 18$. Case 2: First digit is 5 or 7 (2 choices). Last digit has 2 odd choices remaining. Middle 3 digits: $3! = 6$. Total $= 2 \times 2 \times 6 = 24$. Wait, sum $= 18 + 24 = 42$ (or if first is 4, 5, 7). Choices adjusted.

11 A delegation of 4 is chosen from 5 teachers and 6 students. In how many ways can this be done if there must be more teachers than students?

a. **75** ✓

b. 65

c. 85

d. 50

Possibilities: (4 teachers, 0 students) or (3 teachers, 1 student). Case 1: $\binom{5}{4} \times \binom{6}{0} = 5 \times 1 = 5$. Case 2: $\binom{5}{3} \times \binom{6}{1} = 10 \times 6 = 60$. Total $= 5 + 60 = 65$.

12 Four cards are drawn from a standard 52-card deck. What is the probability that all four cards are of the same suit?

a. $\frac{44}{4165} \approx 0.01056$ ✓

b. 0.0025

c. 0.0450

d. 0.0210

Ways to pick 4 cards of one suit $= \binom{13}{4} = 715$. For 4 suits: $4 \times 715 = 2860$. Total 4-card hands $= \binom{52}{4} = 270,725$. Probability $= 2860/270725 = 44/4165 \approx 0.01056$.

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